

# 🌐 HTML NOTES (From Starting to Advanced)

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## 1 What is HTML?

**HTML (HyperText Markup Language)** is used to create the structure of web pages.

- It is **not a programming language**
  - It uses **tags** to define elements
  - It works with **CSS (design)** and **JavaScript (logic)**
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## 2 Basic Structure of HTML

```
<!DOCTYPE html>
<html>
<head>
<title>My First Page</title>
</head>
<body>
<h1>Hello World</h1>
</body>
</html>
```

**Explanation:**

`<!DOCTYPE html>` → tells browser this is HTML5  
`<html>` → root element  
`<head>` → metadata (title, links)  
`<body>` → visible content

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## 3 HTML Tags & Elements

- **Opening tag:** <p>
- **Closing tag:** </p>
- **Element:** <p>Hello</p>

### Types of Tags:

- **Paired Tags** → <p></p>
- **Unpaired Tags** → <br>, <hr>, <img>

## 4 Text Formatting Tags

| Tag          | Use                   |
|--------------|-----------------------|
| <h1> to <h6> | <b>Headings</b>       |
| <p>          | <b>Paragraph</b>      |
| <b>          | <b>Bold</b>           |
| <i>          | <b>Italic</b>         |
| <u>          | <b>Underline</b>      |
| <strong>     | <b>Important text</b> |
| <em>         | <b>Emphasis</b>       |

## 5 Lists in HTML

### Ordered List

```
<ol>
<li>HTML</li>
<li>CSS</li>
</ol>
```

### Unordered List

```
<ul>
<li>Java</li>
<li>Python</li>
</ul>
```

### Description List

```
<dl>
<dt>HTML</dt>
<dd>Markup language</dd>
</dl>
```

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## 6 Links (Anchor Tag)

```
<a href="https://google.com">Go to Google</a>
```

Attributes:

- href → link address
  - target="\_blank" → open in new tab
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## 7 Images

```

```

Attributes:

- src → image path
  - alt → alternate text
  - width / height
- 

## 8 Tables

```
<table border="1">
<tr>
<th>Name</th>
<th>Marks</th>
</tr>
<tr>
<td>Ram</td>
```

```
<td>80</td>
```

```
</tr>
```

```
</table>
```

Tags:

```
<table>, <tr>, <th>, <td>
```

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## 9 Forms (Very Important)

```
<form>
```

Name: `<input type="text"><br>`

Password: `<input type="password"><br>`

```
<input type="submit">
```

```
</form>
```

Input Types:

text, password, email, number

radio, checkbox

submit, reset

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## 10 HTML Attributes

Attributes give extra information.

```
<p align="center">Hello</p>
```

Common Attributes:

- id
  - class
  - style
  - title
- 

## 11 HTML Semantic Tags (Advanced)

Semantic tags give **meaning** to content.

| Tag                          | Use         |
|------------------------------|-------------|
| <code>&lt;header&gt;</code>  | Top section |
| <code>&lt;nav&gt;</code>     | Navigation  |
| <code>&lt;section&gt;</code> | Section     |
| <code>&lt;article&gt;</code> | Article     |
| <code>&lt;aside&gt;</code>   | Sidebar     |
| <code>&lt;footer&gt;</code>  | Footer      |

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## 12 Multimedia Tags

### Audio

```
<audio controls>  
<source src="song.mp3">  
</audio>
```

### Video

```
<video controls width="300">  
<source src="movie.mp4">  
</video>
```

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## 13 HTML Entities

| Entity                  | Symbol |
|-------------------------|--------|
| <code>&amp;lt;</code>   | <      |
| <code>&amp;gt;</code>   | >      |
| <code>&amp;copy;</code> | ©      |

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## 14 Meta Tags (Advanced)

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width,  
initial-scale=1.0">
```

Used for:

- SEO
  - Responsive design
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## **15** HTML5 Features (Advanced)

- Semantic tags
- Audio / Video support
- Local Storage
- Form validation